



Sherlocks

Baggage - Very Easy

WRITE-UP

Lesson Learned

- How to use tool Eric Zimmerman (ShellBags Explorer)
- Know the windows registry process about ShellBags
- Malicious extraction tool name
- Favourite place of hacker to persistence in windows files

Sherlock Scenario

This Sherlock provides players with an opportunity to analyze Shellbag artifacts. Shellbags can be used to find evidence of folder access by a specific user, access to network shares, and navigation of archive file contents. This information can be leveraged during investigations to identify potential data access, data staging, and data exfiltration attempts.

Must to know:

- **ShellBags Explorer** is a free application used by computer forensic experts to view the history of folders that users have opened or modified on the Windows operating system.
- **ShellBags** are a set of secret entries in the **Windows Registry** that serve to remember the display settings for every folder you've ever opened in Windows Explorer.
- A **credential extraction tool** is software or a digital tool used to retrieve sensitive data—such as usernames, passwords, and access tokens—stored within a computer system or a device's memory.
- The **Temp** folder is a favorite spot for storing malicious files. Why? Because it's a temporary location, which allows malicious files to blend in with other files; it has lax permissions; and it's rarely checked manually. There are many other favorite locations that meet the criteria above.

Important tree to investigate

```
Baggage.zip
├── Baggage
│   └── C
│       └── Users
│           └── steve
│               └── NTUSER.DAT
│               ├── ntuser.dat.LOG1
│               ├── ntuser.dat.LOG2
│               └── AppData
│                   └── Local
│                       └── Microsoft
│                           └── Windows
│                               ├── UsrClass.dat
│                               ├── UsrClass.dat.LOG1
│                               └── UsrClass.dat.LOG2
```

← USER TARGET

← ★ SHELLBAGS

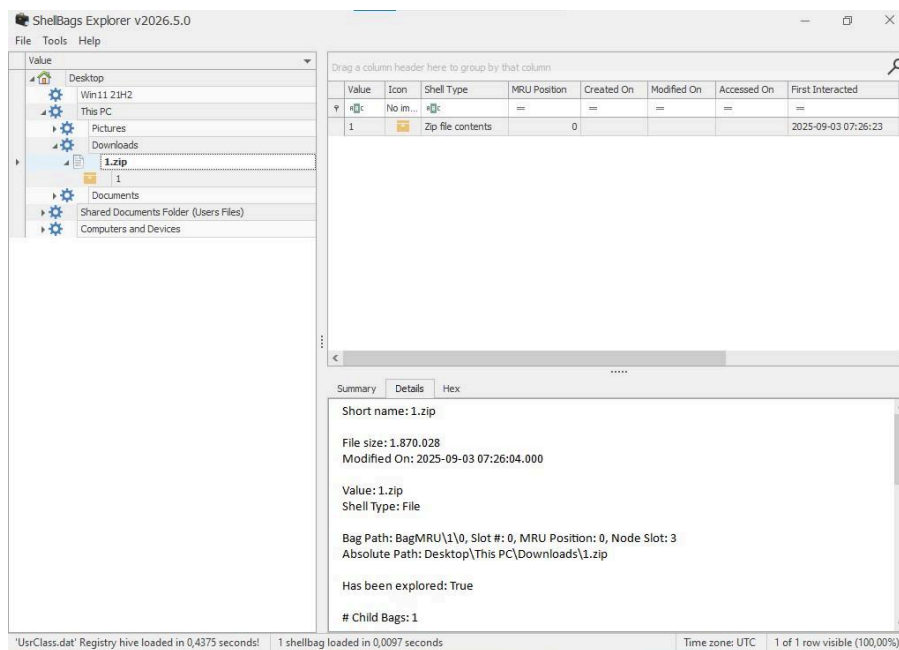
In this case the victim is steve users, so without any further a do let's jump in!

Task 1

What was the name of the archive file downloaded by the compromised account?

For the first step to answer this task, we just need to check the Downloads folder, open file in Eric Zimmerman tool ShellBags Explorer

Go To : Desktop → This PC → Downloads → the archive file



In Downloads folder there just one file archive and that's the answer we found

Answer = 1.zip

Task 2

What was the name of the utility brought in by the attacker to search for sensitive data?

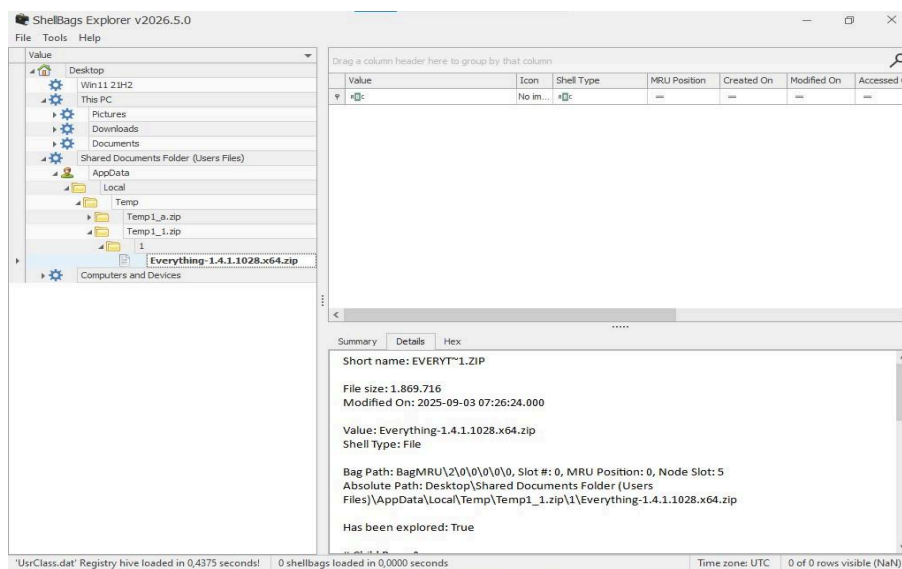
To find this we must know the tool likely for searching sensitive data or extraction, i found this list from internet:

- Everything (an extremely fast instant file search utility, which is often misused by attackers to track down sensitive documents).
- LaZagne / Mimikatz (tools for finding and collecting credentials).
- Advanced IP Scanner / Angry IP Scanner (for finding sensitive devices on a network).

We just digging to find related named in the list, and this the path to find this utility

Go To : Desktop → Shared Documents Folder (Users Files) → AppData → Local → Temp → Temp1_1.zip → 1 → **Everything-1.4.1.1028.x64.zip**

This Temp folder likely place for attacker to place the malicious file, and we got the answer adjust with format answer in hackthebox



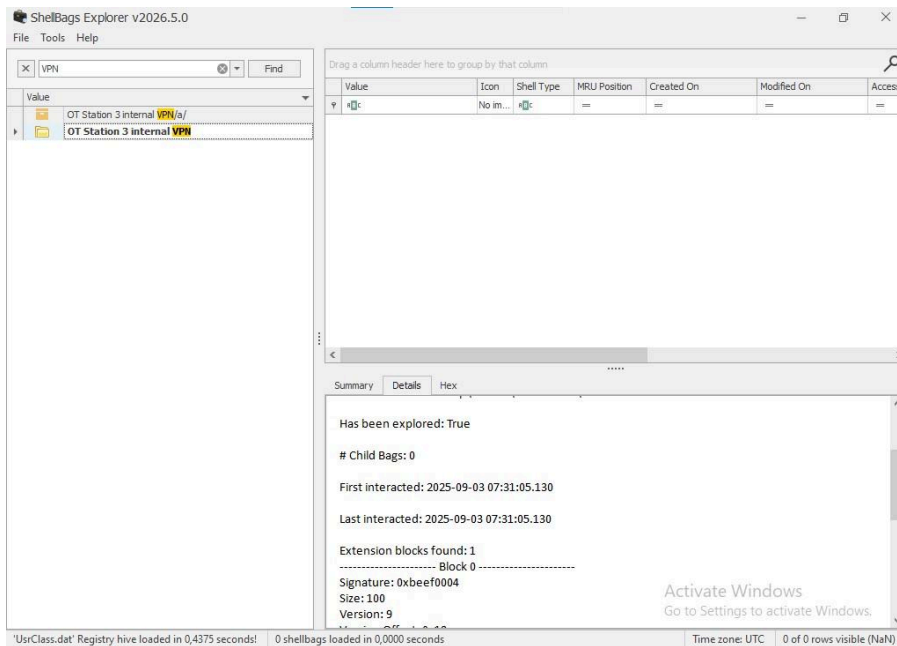
Answer = Everything 1.4.1.1028

Task 3

Hint : Look for Last Interacted Timestamp for the VPN directory.

The attacker navigated the filesystem and found sensitive files used by the victim in their day-to-day work. When was the VPN folder accessed by the attacker?

Task 3 we must look directory likely VPN as a named, using CTRL + F and type “VPN” we find this folder related to this task OT Station 3 internal VPN and now we can read the details to find Last Interacted Timestamp!



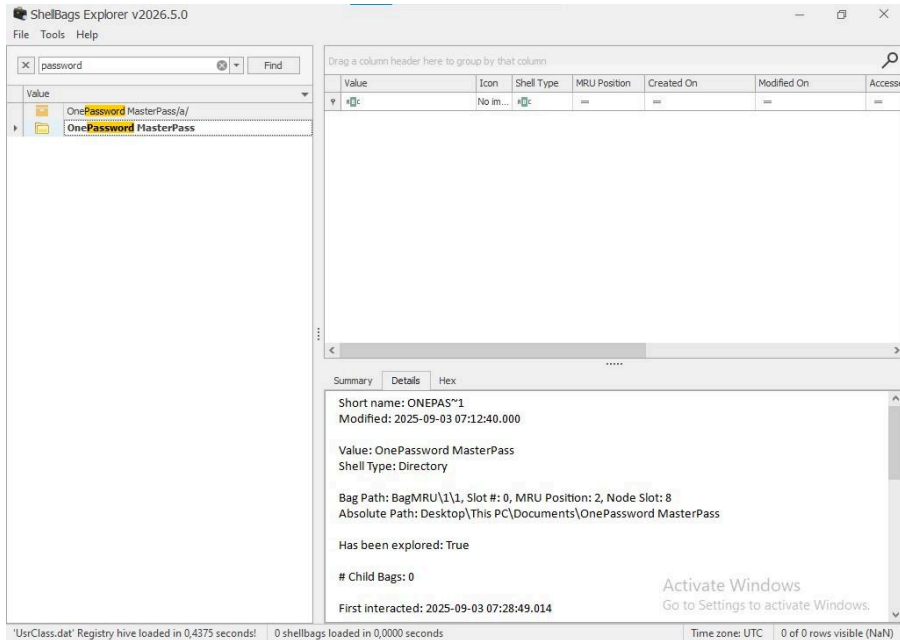
we can see in this line and found it the answer : Last interacted: 2025-09-03 07:31:05.130 and adjust for the formatting

Answer = 2025-09-03 07:31:05

Task 4

What was the name of the directory containing the victim's passwords?

Using CTRL + F again and type “password” we can find the victim directory related for this task

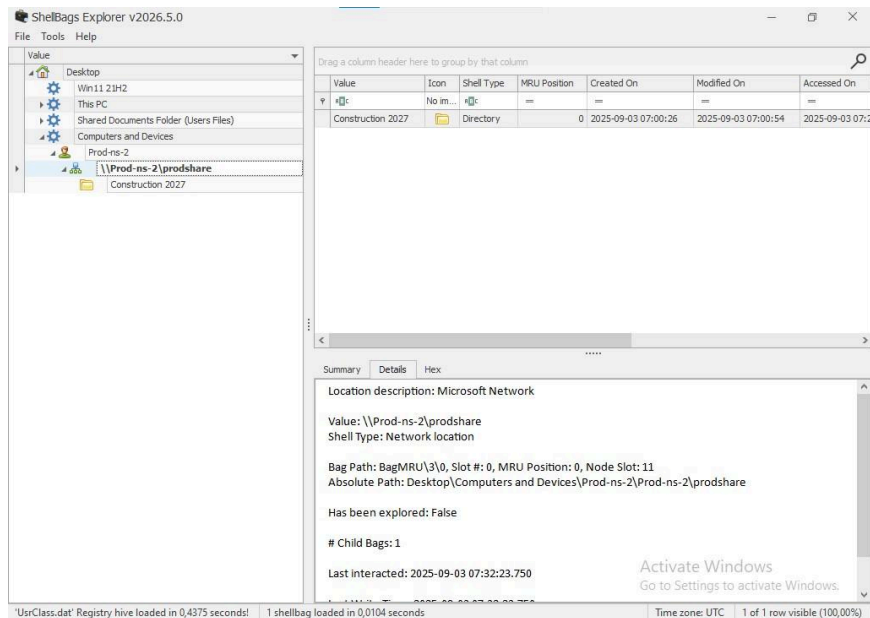


Answer = OnePassword MasterPass

Task 5

The attacker also accessed a network share to pillage network data. What is the UNC path?

The folder related for this task is Computers and Devices so i go there digging to the tree and found some juicy UNC Path

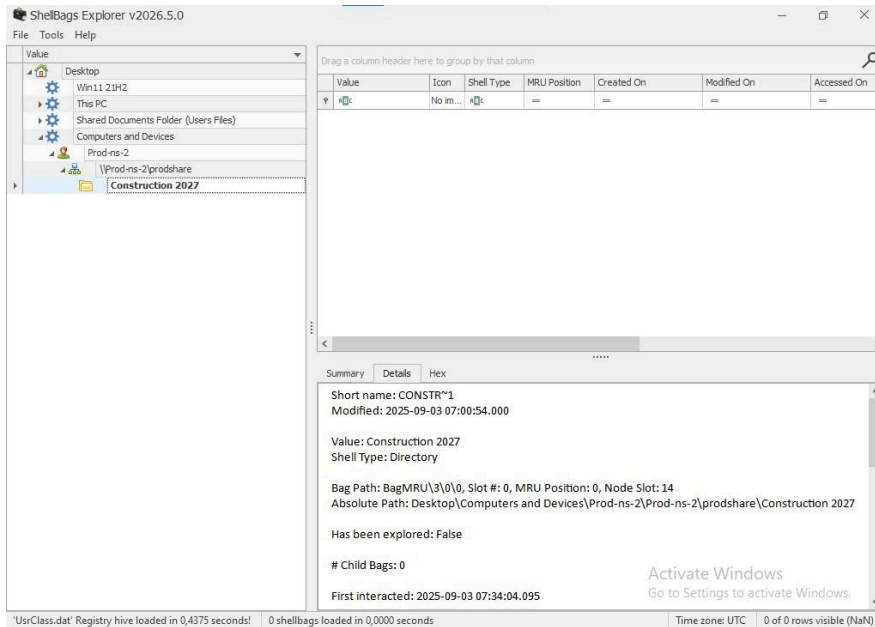


Answer = \\Prod-ns-2\\prodshare

Task 6

When is the dam construction planned?

In the same directory in previous task we can find the folder related for construction the year show for the answer

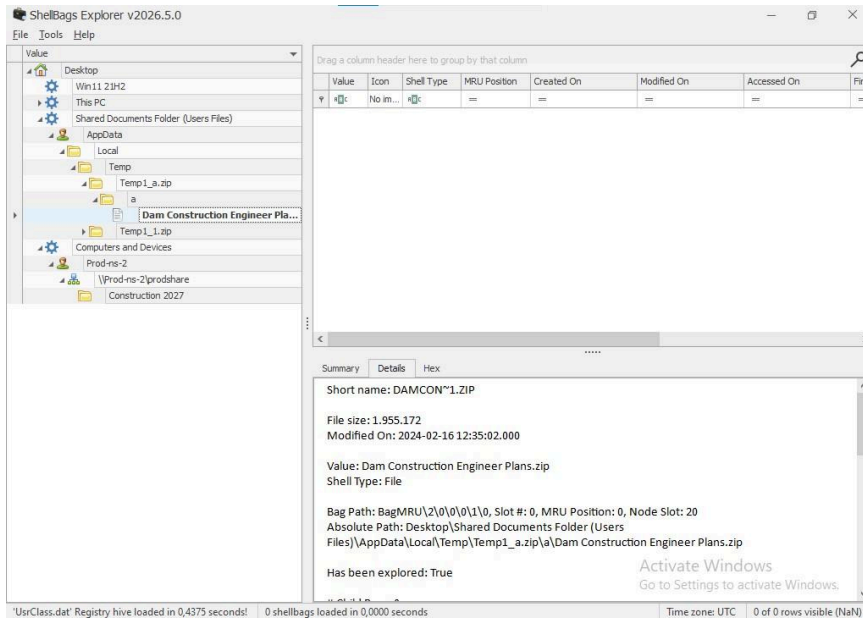


Answer = 2027

Task 7

What was the name of the archive file present on the network share?

Go to this path and found the archive file : Desktop\Shared Documents Folder (Users Files)\AppData\Local\Temp\Temp1_a.zip\

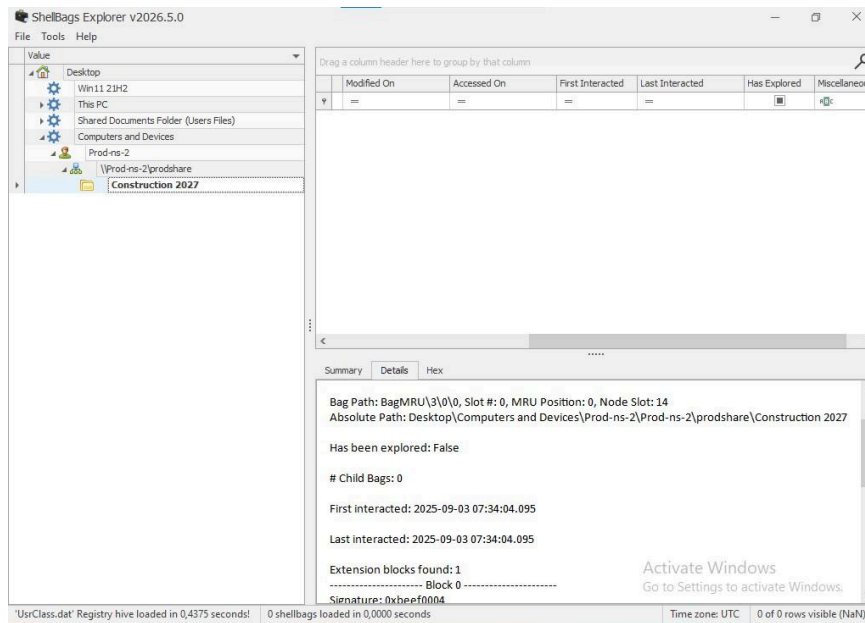


Answer = Dam Construction Engineer Plans.zip

Task 8

When was the archive file from the network share accessed?

Check the folder for network share in previous task 6 we find the folder named Construction 2027 and see the details for Interacted Timestamp



Answer = 2025-09-03 07:34:04

The attacker created a staging folder to prepare for collection and exfiltration. What is the full path of the staging folder?

We find the interesting folder in Pictures directory the attacker prepare for the extraction in this directory because this file executable the folder named is a there are three folder for the extraction : OT Station 3 internal VPN, OnePassword MasterPass, Engineers Tab. From the ROOT tree for archive given in this case we know the path to this directory!

ShellBags Explorer v2026.5.0

File Tools Help

Value

- Desktop
 - Win 11 21H2
 - This PC
 - Pictures
 - a.zip
 - a
 - OT Station 3 internal VPN/a/
 - OnePassword MasterPass/a/
 - Engineers Tab/a/
 - a
 - Downloads
 - Documents
 - Shared Documents Folder (Users Files)
 - Computers and Devices

Drag a column header here to group by that column

| n | Created On | Modified On | Accessed On | First Interacted | Last Interacted | Has Explored | Miscellaneous |
|---|------------|-------------|-------------|---------------------|---------------------|-------------------------------------|---------------|
| 2 | | | | 2025-09-03 07:34:34 | | ☒ | |
| 1 | | | | 2025-09-03 07:34:35 | | ☒ | |
| 0 | | | | 2025-09-03 07:34:37 | 2025-09-03 07:34:37 | ☒ | |

Summary Details Hex

Last access internal value:

Value: a

Shell Type: Zip file contents

Bag Path: BagMRU\1\2\1, Slot #: 0, MRU Position: 0, Node Slot: 16

Absolute Path: Desktop\This PC\Pictures\1a.zip\1a

Has been explored: True

Child Bags: 3

Last Interacted: 2025-09-03 07:34:32.626

Activate Windows
Go to Settings to activate Windows.

'UserClass.dat' Registry file loaded in 0,4375 seconds! 3 shellbags loaded in 0,0051 seconds

Time zone: UTC 3 of 3 rows visible (100,00%)

Answer = C:\users\Steve\Pictures

Task 10

The attacker compressed the staging folder to prepare the data for exfiltration. When was the exfiltration archive file accessed?

In same directory we can find the correlation after the attacker prepare staging folder the next step he used is compressed the folder into archive a.zip file and we can see the details to answer this task in line Last interacted: 2025-09-03 07:34:30.081

The screenshot shows the ShellBags Explorer v2026.5.0 interface. The left pane displays a file tree with 'a.zip' selected under 'Pictures'. The right pane shows a table with columns: Created On, Modified On, Accessed On, First Interacted, Last Interacted, Has Explored, and Miscellaneous. The table contains one row with the following data:

| Created On | Modified On | Accessed On | First Interacted | Last Interacted | Has Explored | Miscellaneous |
|------------|-------------|-------------|------------------|---------------------|-------------------------------------|---------------|
| | | | | 2025-09-03 07:34:32 | ☒ | |

Below the table, the 'Details' tab is active, showing the following information:

- Bag Path: Bag\MRU\1\2, Slot #: 1, MRU Position: 0, Node Slot: 15
- Absolute Path: Desktop\This PC\Pictures\a.zip
- Has been explored: True
- # Child Bags: 1
- Last interacted: 2025-09-03 07:34:30.081
- Extension blocks found: 1
- Block 0 -----
- Signature: 0xbeef0004
- Size: 60
- Version: 9

The status bar at the bottom indicates: 'UsrClass.dat' Registry hive loaded in 0,4375 seconds! 1 shellbag loaded in 0,0034 seconds Time zone: UTC 1 of 1 row visible (100,00%)

Answer = 2025-09-03 07:34:30

Bravo!!! Case Close!!!